

SUNREY

POST-QUANTUM SECURITY ARCHITECTURE

A Crypto-Agile, Zero-Trust Security Architecture for Financial Infrastructure, Blockchain and the AI Economy

CORE SECURITY THESIS

SunRey is designed so that no single cryptographic key, AI model, application service, blockchain node, external provider, database, employee or network connection becomes unrestricted economic authority.

Cryptography protects communication and state. Deterministic authorization protects economic authority. Evidence proves what occurred.

Technical White Paper | Version 1.0

September 2026 | SunRey

Abstract

SunRey is an AI-native financial and economic operating system whose security problem spans more than confidentiality. It must protect money movement, AI-initiated actions, blockchain state, custody, personal information, market infrastructure, provider connections, machine identities, governance and the software supply chain without allowing any one security mechanism to become a universal authority. This paper describes a layered architecture in which post-quantum readiness, cryptographic agility, zero-trust access, deterministic authorization, canonical ledgers, sovereign blockchain state, key-purpose separation, protected data domains, evidence-producing execution and operational resilience operate as one security fabric.

The architecture is deliberately more conservative than the phrase "quantum-proof." SunRey source material describes an algorithm registry, security KeyProvider abstractions, hybrid post-quantum migration concepts, ML-KEM/ML-DSA/SLH-DSA testnet-oriented ports, signer safety, HSM/KMS interfaces, zero-trust controls and external assurance gates. Those elements support a credible post-quantum migration program, but they do not establish that every production communication, wallet, validator or custody path is already post-quantum protected. [S4]

CLAIM DISCIPLINE

This paper distinguishes architecture defined, code implemented, sandbox/testnet qualification, production configuration, external assurance and production activation. No layer is described as certified or "quantum-proof" without evidence.

Publication boundary

This document is a public architecture paper. It intentionally excludes credentials, private network addressing, sensitive administrative endpoints, firewall specifics, recovery secrets, key ceremony secrets, provider credentials and confidential vulnerability findings. It is not a penetration-test report, certification, legal opinion or production-security attestation.

External standards check

The standards references in this edition were checked against current NIST, CISA, NSA and IETF materials available in September 2026. NIST FIPS 203, 204 and 205 standardize ML-KEM, ML-DSA and SLH-DSA; NIST SP 800-227 provides KEM guidance; NIST CSWP 39upd1 addresses crypto agility; and IETF RFC 10024, published in August 2026, defines hybrid TLS 1.3 key agreements combining ML-KEM with ECDHE. [E1-E8]

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READING GUIDE

Sections 1-4 establish the trust model. Sections 5-10 define cryptographic and post-quantum architecture. Sections 11-23 apply security to the blockchain, AI, data, providers, custody, Exchange and infrastructure. Sections 24-30 address assurance, operations and migration.

1 | EXECUTIVE SUMMARY

Executive Summary

The objective is not merely stronger encryption; it is controlled economic authority under changing cryptographic assumptions.

SunRey combines financial accounts, global money movement, AI financial agents, a proprietary blockchain, SunRey Coin, MoonRey Coin, an economic Exchange, protected human-information rights, wallets, custody, oracles and productive-capacity markets. That combination creates a security perimeter that is broader than the perimeter of either a conventional bank application or a standalone blockchain. A compromise can originate at the user device, an AI model, an API, a provider, a validator, a software dependency, a key-management system, a database, an oracle or a governance process. The architecture therefore cannot rely on one perimeter, one signature system or one notion of trust.

The central design rule is separation of authority. SunRey source architecture requires consequential actions to flow through an authorization spine: ActionIntent -> Compliance Kernel -> signed Execution Authority -> Financial Fabric -> canonical Ledger or blockchain settlement -> Evidence Vault. AI agents may discover and propose; policy determines permission; deterministic risk controls determine exposure; controlled infrastructure executes; canonical state records the result. [S1][S2][S4]

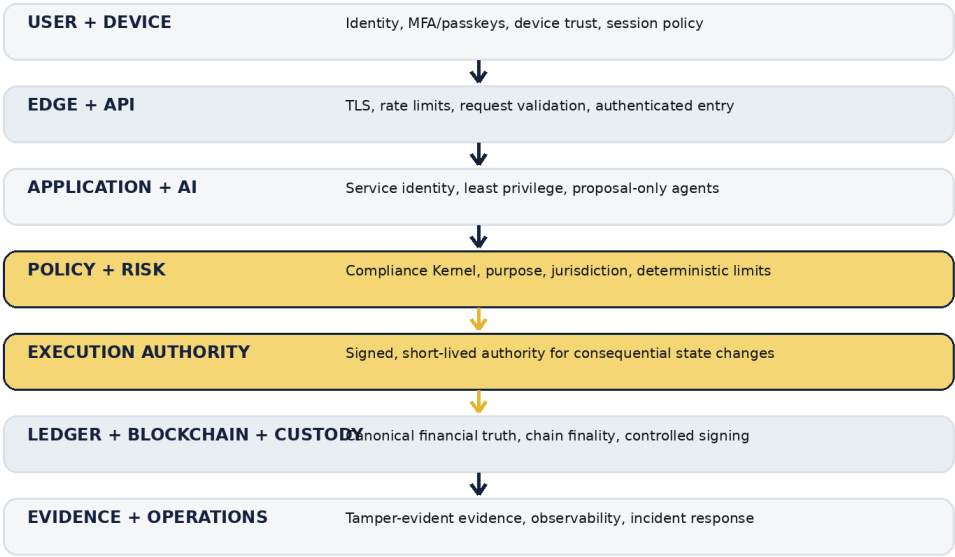
SECURITY INVARIANT

Compromise of one layer should not automatically compromise economic authority in another. A stolen API credential should not become a signing key. A model compromise should not become a wallet compromise. A valid signature should not automatically mean a transaction is policy-authorized.

Post-quantum readiness is integrated into this model as a transition discipline. The SunRey architecture reports algorithm registries, provider ports, distinct key purposes, hybrid migration concepts and testnet-oriented ML-KEM/ML-DSA/SLH-DSA integration points. The right public claim is therefore crypto-agile and post-quantum-ready by architecture, with production use gated by qualification, provider support and independent review. [S4]

SunRey Defense-in-Depth Architecture

Security controls are layered so compromise of one plane does not automatically confer economic authority.



Design principle: authentication is not authorization; authorization is not unrestricted execution.

Figure 1. SunRey defense-in-depth architecture. Cryptography is one layer inside a larger authority and evidence model.

2 | SECURITY THESIS AND CLAIM DISCIPLINE

Security Thesis and Claim Discipline

A security white paper should distinguish what exists in architecture from what is operationally proven.

2.1 Four security questions

- Can an actor or service prove who or what it is?
- Is that actor authorized for this specific action, purpose and context?
- Can the system preserve confidentiality and authenticity under current and future cryptographic assumptions?
- Can the platform reconstruct what happened and recover without creating a second source of financial truth?

Traditional cybersecurity often stops after identity and cryptography. SunRey adds a separate economic-authority layer because authenticated systems can still be wrong, compromised, misconfigured or legally unauthorized. This matters particularly for autonomous agents and external providers: a correct TLS session proves a secure channel to an endpoint; it does not prove that the provider response should create money, move funds, change a user's risk mandate or alter protocol governance.

2.2 Maturity states

State	Meaning	Permitted public wording
Architecture defined	Controls, interfaces and authority boundaries are specified.	"Designed for" / "architecture supports"
Implemented in code	A control exists in the repository.	"Implemented component" when verified
Sandbox / testnet qualified	The control has passed bounded non-production tests.	"Qualified in controlled environment"
Production configured	Real keys, providers and infrastructure are bound.	"Production-configured" only with evidence
Externally assured	Independent review/audit evidence exists.	Name the scope and reviewer; avoid blanket claims
Production active	The exact control is operating in the live service.	Scope claim to the specific path and version

PUBLIC CLAIM RULE

Never compress these states into a single statement such as "SunRey is quantum-proof" or "SunRey is fully post-quantum encrypted."

3 | THREAT MODEL FOR A POST-AI ECONOMIC SYSTEM

Threat Model for a Post-AI Economic System

The attack surface includes both cyber compromise and authority confusion.

SunRey must defend not only against classic confidentiality, integrity and availability failures, but also against an attacker causing one subsystem to be accepted as an authority it was never meant to hold. The highest-risk failures therefore include both direct compromise and cross-plane privilege escalation.

Threat family	Example attack	Primary authority at risk
Cryptographic	Private-key theft, downgrade, weak or deprecated algorithm	Confidentiality, signatures, service identity
Identity	Credential theft, device takeover, impersonation	User or operator identity
AI	Prompt injection, malicious tool call, model compromise	Proposal and decision-support plane
Financial	Unauthorized payment, duplicate settlement, ledger manipulation	Canonical financial state
Blockchain	Validator key theft, double-signing, consensus fault	Protocol finality and native assets
Wallet / custody	Unauthorized signing or recovery abuse	Digital-asset control
Exchange	Order abuse, market manipulation, settlement inconsistency	Market integrity and custody linkage
Data	Exfiltration or purpose expansion	Human information rights and privacy
Oracle	Spoofed, stale or duplicated economic facts	External-fact quality
Provider	Compromised bank/KYC/market-data/API dependency	External service trust
Supply chain	Compromised dependency, build or release artifact	Software integrity
Governance	Captured approval, malicious upgrade or key misuse	Protocol/control-plane authority

Security Control Chain

Map every material threat to prevention, detection, authority boundaries, recovery and evidence.

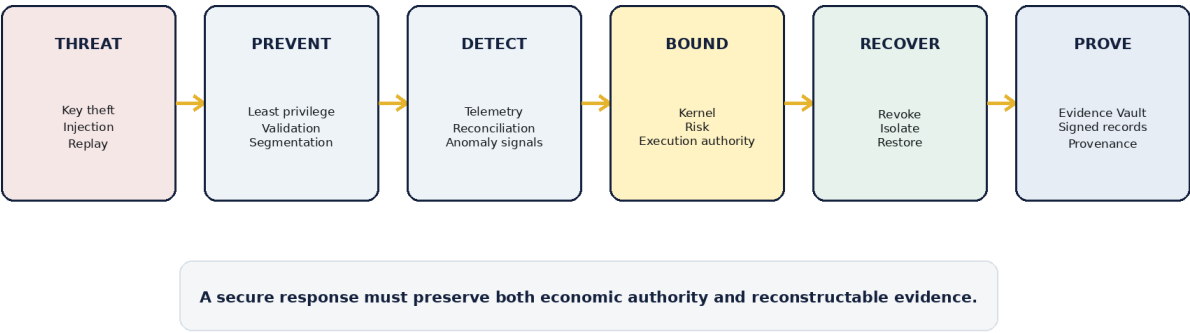


Figure 2. A threat-control-evidence chain suitable for financial infrastructure.

4 | DEFENSE-IN-DEPTH SECURITY ARCHITECTURE

Defense-in-Depth Security Architecture

Security controls are layered around identity, policy, execution and evidence rather than around a single perimeter.

The SunRey security model is consistent with zero-trust architecture: internal network location is not itself a reason to trust a user, workload or service. NIST SP 800-207 similarly defines zero trust around removing implicit trust based on network location and protecting resources through explicit authentication and authorization. [E9]

4.1 Security planes

- User and device plane - identity verification, MFA/passkeys, trusted-device state, session controls and step-up authentication.
- Edge and API plane - TLS, API authentication, request validation, idempotency, rate limiting, replay controls and browser-origin policy.
- Application and AI plane - workload identity, least privilege, capability tokens, model/tool restrictions and bounded proposal generation.
- Policy and risk plane - Compliance Kernel, jurisdiction packs, purpose controls, deterministic risk limits and kill switches.
- Execution plane - signed, short-lived Execution Authority as the gate to consequential state change.
- State plane - canonical financial Ledger, SunRey Blockchain finality, custody and immutable/reconstructable evidence.
- Operations plane - observability, anomaly detection, incident response, backup/restore, security review and release governance.

DEFENSE-IN-DEPTH PRINCIPLE

Security mechanisms should fail independently. If one control weakens, downstream controls should still constrain the blast radius and preserve evidence.

5 | QUANTUM RISK AND LONG-LIVED FINANCIAL DATA

Quantum Risk and Long-Lived Financial Data

The most immediate PQC problem is migration planning, not speculation about the date of a cryptographically relevant quantum computer.

Current public-key systems used for key establishment and digital signatures are the primary migration concern for a future cryptanalytically relevant quantum computer. CISA, NSA and NIST have urged organizations to plan early because migration is large, dependencies are difficult to inventory, and data captured today may retain value long enough to be attacked later - the "harvest now, decrypt later" problem. [E6]

Financial systems contain unusually long-lived assets: identity records, legal agreements, transaction evidence, custody records, private economic data, audit histories and institutional credentials. Even if a future cryptographic break is years away, a system that cannot locate its quantum-vulnerable dependencies cannot make a controlled transition when required.

5.1 SunRey cryptographic exposure classes

Exposure class	Examples	Migration priority
Long-lived confidentiality	Vault exports, archived financial evidence, backups, sensitive institutional records	Prioritize discovery and protection strategy
Long-lived authenticity	Governance attestations, release provenance, archival signatures	Define re-signing / renewal / evidence policy
Transactional authentication	TLS, service identity, provider API authentication	Crypto-agile protocol and vendor readiness
Blockchain signatures	Wallet, validator, governance and attestation signatures	Versioned protocol transition; avoid uncoordinated changes
Third-party dependency	Banks, custody, cloud, HSM/KMS, mobile platforms	Vendor roadmap and acceptance evidence

WHY PREPARE NOW

The security objective is to reduce migration surprise. Inventory, abstraction, versioning and vendor coordination can be implemented before production deployment of any specific PQ algorithm.

6 | CRYPTO AGILITY AS AN ARCHITECTURAL REQUIREMENT

Crypto Agility as an Architectural Requirement

Post-quantum security is sustainable only if algorithms can change without redesigning the economic system.

NIST defines crypto agility as the capabilities needed to replace and adapt cryptographic algorithms across protocols, applications, software, hardware, firmware and infrastructure while preserving security and ongoing operations. Its updated 2026 guidance emphasizes operational mechanisms and trade-offs rather than a single universal implementation pattern. [E5]

SunRey source architecture aligns with this direction by describing an algorithm registry, provider ports, established primitives, hybrid post-quantum migration and a ban on homegrown cryptography. [S4] The architectural goal is that financial, blockchain and data semantics remain stable while cryptographic suites can be versioned, qualified and replaced.

6.1 Algorithm Registry

Registry field	Purpose
Algorithm / suite identifier	Canonical reference used by protocol and evidence
Algorithm family	KEM, digital signature, symmetric encryption, hash, MAC, key derivation
Lifecycle state	Research, qualified, approved, deprecated, disabled
Environment scope	Development, testnet, sandbox, preproduction, production
Key purpose	Service identity, wallet, validator, governance, encryption, release signing
Provider / module	Software library, HSM, KMS, custody signer or platform implementation
Version / parameters	Prevents ambiguous or silent changes
Compatibility profile	Protocol and external-party requirements
Evidence links	Test vectors, assurance results, approval records

NO HOMEGROWN CRYPTOGRAPHY

SunRey may innovate in economic state, authorization and protocol composition; cryptographic primitives should come from established, reviewed standards and implementations.

7 | POST-QUANTUM STANDARDS LANDSCAPE - 2026

Post-Quantum Standards Landscape - 2026

Standards are mature enough to plan against, but deployment choices still require system-level engineering and assurance.

NIST approved the first three primary PQC FIPS standards in August 2024. FIPS 203 specifies ML-KEM for key establishment; FIPS 204 specifies ML-DSA for digital signatures; and FIPS 205 specifies SLH-DSA, a stateless hash-based signature scheme. [E1-E3] NIST SP 800-227, finalized in September 2025, provides recommendations for KEM use. [E4]

Standard	Role	SunRey relevance
FIPS 203 - ML-KEM	Post-quantum key encapsulation	Candidate foundation for qualified key-establishment paths; not proof of current production use
FIPS 204 - ML-DSA	Post-quantum digital signatures	Candidate for versioned signing roles subject to performance, HSM and protocol qualification
FIPS 205 - SLH-DSA	Stateless hash-based signatures	Alternative signature family with different performance/size characteristics
SP 800-227	Secure KEM implementation guidance	Useful for implementation and combiner design review
CSWP 39upd1	Crypto-agility strategies and practices	Supports algorithm abstraction, inventory and transition discipline
RFC 10024	Hybrid PQ/traditional TLS 1.3 key agreements	Relevant to external/internal transport migration and compatibility testing

IETF RFC 10024 became a Standards Track RFC in August 2026. It defines TLS 1.3 hybrid groups that combine ML-KEM with ECDHE, including X25519MLKEM768 and NIST-curve variants. This is directly relevant to SunRey's hybrid-transport strategy because it provides a standards-based path for combining conventional and post-quantum key agreement during migration. [E8]

STANDARD AVAILABLE != PRODUCTION DEPLOYED

The existence of a standard does not prove that a particular SunRey service, HSM, wallet, validator or provider has implemented and qualified it.

8 | DATA IN TRANSIT AND HYBRID TLS

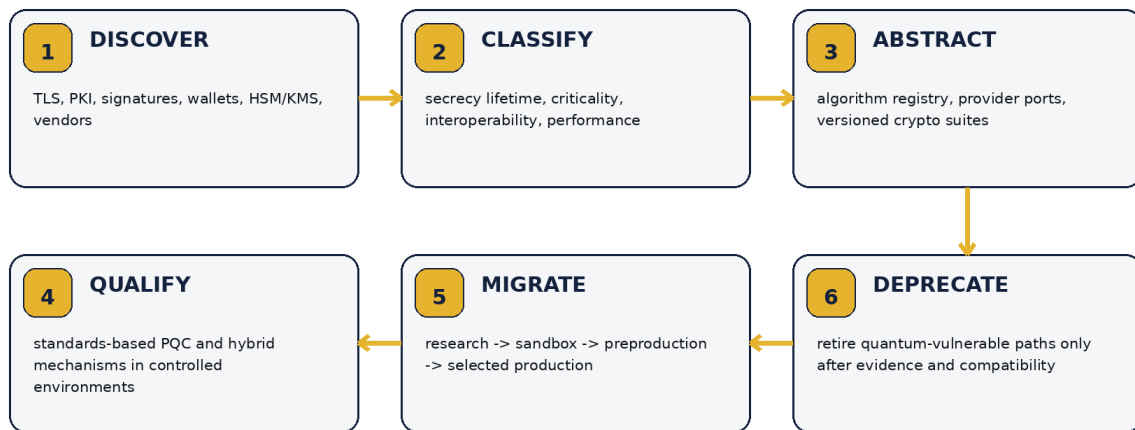
Data in Transit and Hybrid TLS

Transport protection must evolve without breaking provider interoperability or weakening identity and authorization.

SunRey depends on many communications paths: mobile/web clients to API edge; service-to-service calls; provider adapters; custody signers; market-data feeds; oracle connections; validator/P2P links; operator access and external interoperability. Each path should be inventoried by protocol, endpoint class, certificate/signature dependency, confidentiality lifetime and external compatibility requirement.

Post-Quantum Migration Architecture

SunRey treats PQC as a governed transition across cryptographic dependencies, not a one-time algorithm swap.



Target property: preserve confidentiality, authenticity, authorization and operational continuity while cryptographic assumptions evolve.

Figure 3. Post-quantum migration should proceed as a governed lifecycle across cryptographic dependencies.

8.1 Hybrid migration

Hybrid key agreement is useful where organizations want protection that does not depend solely on a new PQ primitive while maintaining transition compatibility. RFC 10024 standardizes specific PQ/traditional hybrid TLS 1.3 groups. SunRey can evaluate these or successor standards within a qualified transport profile, measuring handshake size, latency, CPU cost, certificate/tooling compatibility, provider support and failure behavior. [E8]

8.2 Transport rules

- No cleartext internal assumption merely because traffic is "inside" a cloud or VPC.
- Mutual or workload-authenticated channels where service identity is consequential.
- Provider endpoints treated as external evidence sources even when TLS authentication succeeds.
- Protocol downgrade and legacy-suite use observable and policy-controlled.
- Transport configuration versioned and associated with deployment evidence.

9 | DATA AT REST, SECRETS AND PROTECTED STORAGE

Data at Rest, Secrets and Protected Storage

Encryption at rest is necessary but insufficient; key separation, access policy and purpose limitation determine the real trust boundary.

The SunRey architecture places particularly sensitive human source data outside the public blockchain and inside protected storage domains. The original architecture also calls for independently encrypted Vault categories, separate encryption keys where appropriate, a Purpose Firewall and a revocable Consent Ledger. [S2] This reduces the risk that economic coordination requires permanent publication of raw personal data.

9.1 Protected storage classes

Data class	Security objective	Preferred control pattern
Financial ledger / account state	Integrity, recoverability, access control	Encrypted databases, role-separated access, immutable journal semantics, backup/restore
Human information source data	Confidentiality, purpose limitation, revocation	Segmented Vault, independent keys, policy enforcement, clean-room compute
Audit / evidence	Integrity and reconstruction	Tamper-evident or WORM-oriented storage, signed/hash-linked records
Provider credentials	Confidentiality and rotation	Secret references, managed secret stores, least privilege, no source-code secrets
Backups	Confidentiality and availability	Encrypted, access-separated, tested restore, off-system copies where appropriate

DATA RULE

Encryption answers "who can read the bytes?" Purpose controls answer "who may use the information, for what purpose, and for how long?" SunRey needs both.

10 | KEY MANAGEMENT AND ROOTS OF TRUST

Key Management and Roots of Trust

Key material should be purpose-separated, lifecycle-managed and incapable of silently becoming universal authority.

SunRey source architecture describes a Security KeyProvider/root security abstraction and later HSM/KMS signer ports, approval controls, recovery/rotation/delegated keys and production key-management evidence as a launch gate. [S4][S6] This is the correct decomposition: applications should depend on controlled signing/encryption interfaces rather than directly handling unrestricted key material.

Key Purpose Separation and Root-of-Trust Architecture

Cryptographic possession proves control of a key; policy determines what that key is authorized to do.

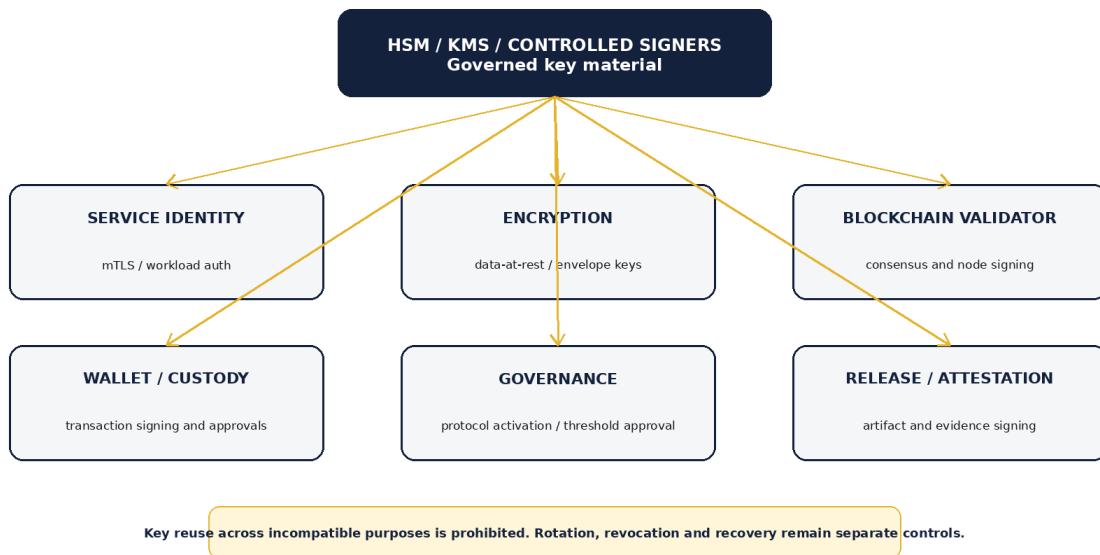


Figure 4. Key-purpose separation prevents a single credential from becoming a universal security root.

10.1 Key lifecycle

1. Generate or import using an approved provider and documented entropy/source requirements.
2. Bind the key to a purpose, environment, owner/controller, algorithm suite and policy state.
3. Restrict use through service identity, human approval, rate/amount constraints or threshold control as appropriate.
4. Rotate and revoke without rewriting the economic object model.
5. Preserve auditable evidence of creation, use, rotation, compromise response and retirement.

KEY POSSESSION != ECONOMIC AUTHORITY

A cryptographically valid signature proves that a key was used. A separate policy layer should still determine whether that key was permitted to authorize the requested economic state change.

11 | SUNREY BLOCKCHAIN SECURITY

SunRey Blockchain Security

A sovereign economic state machine has to protect actors, rights, policy and evidence - not only token balances.

SunRey Blockchain is designed as the sovereign base layer for native SunRey economic objects, SunRey Coin, MoonRey Coin and core protocol state. The architecture treats human, enterprise, AI-agent, robot, device and productive-asset identities; rights and consent; provenance; policy/jurisdiction; productive capacity; settlement and evidence as first-class protocol concepts. [S5]

SunRey Blockchain Security Domains

Security extends beyond ledger integrity to actors, rights, policy, evidence and governance.

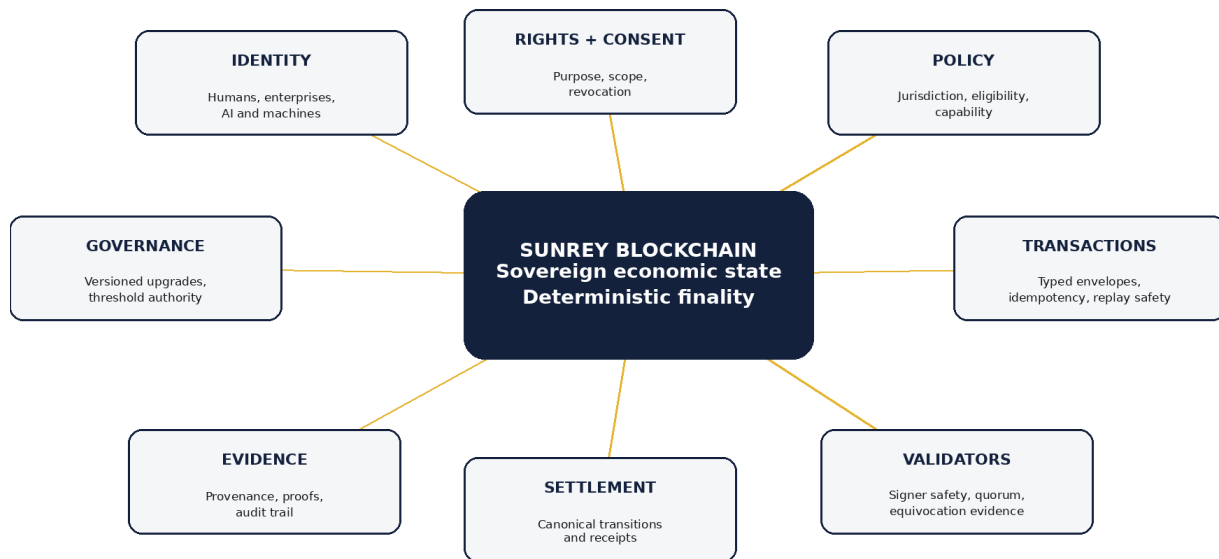


Figure 5. Security domains within the SunRey sovereign economic state machine.

11.1 Deterministic execution and canonical state

A security-sensitive execution environment should favor deterministic state transitions, canonical encoding, explicit resource accounting, versioned transaction envelopes and reproducible validation. Source build reports describe canonical binary encoding, no floating-point consensus values, deterministic execution and BFT-oriented finality work. [S4] These controls reduce cross-implementation ambiguity and make replay, hashing and signing behavior tractable.

11.2 Rich transaction security

Because SunRey transactions can carry identity, rights, purpose, jurisdiction, consent references, oracle state and evidence requirements, protocol validation can reject transactions that are cryptographically valid but contextually unauthorized. This is an important distinction from a narrow account-balance model. [S5]

12 | VALIDATOR, P2P AND CONSENSUS SECURITY

Validator, P2P and Consensus Security

Validator safety depends on keys, operator identity, network topology, accountability and controlled upgrades.

The source architecture describes validator operators as bonded named principals, separate consensus and P2P keys, epoch-based set changes, equivocation evidence, authenticated node identities, sentry-capable topology, consensus-priority traffic and RPC separation. It also reports signer fencing, double-sign protection and BFT-style rounds in the testnet/protocol program. [S4]

12.1 Validator threat controls

Threat	Control objective	Evidence
Validator key theft	HSM/KMS or controlled signer, key-purpose separation, rotation/revocation	Signer logs, key state, incident record
Double signing / equivocation	Signer fencing, height/round safety, evidence and penalties/governance response	Consensus evidence
P2P DoS	Sentry topology, peer policy, RPC separation, resource controls	Network telemetry and alerts
Malicious upgrade	Versioned governance, activation coordinates, threshold human authority	Signed governance package and release provenance
Consensus divergence	Deterministic encoding/execution and bounded protocol versions	Cross-node test vectors and replay evidence

GOVERNANCE BOUNDARY

A software release signature or AI recommendation should not itself authorize a protocol upgrade. Upgrade authority belongs to explicit governance.

13 | TRANSACTION SECURITY AND PROOF OF PERMISSION

Transaction Security and Proof of Permission

Every consequential state change should carry both cryptographic validity and contextual authorization.

SunRey's Proof of Permission model requires six authorization classes - Identity, Authority, Jurisdiction, Compliance, Risk and Purpose - before the Compliance Kernel may issue an Execution Authority. The authorization posture is monotonic: a request may become more restrictive as evidence changes, but should not silently relax mid-flight. [S1]

AI Authority Separation

AI can research and propose, but deterministic systems retain permission, risk and execution authority.

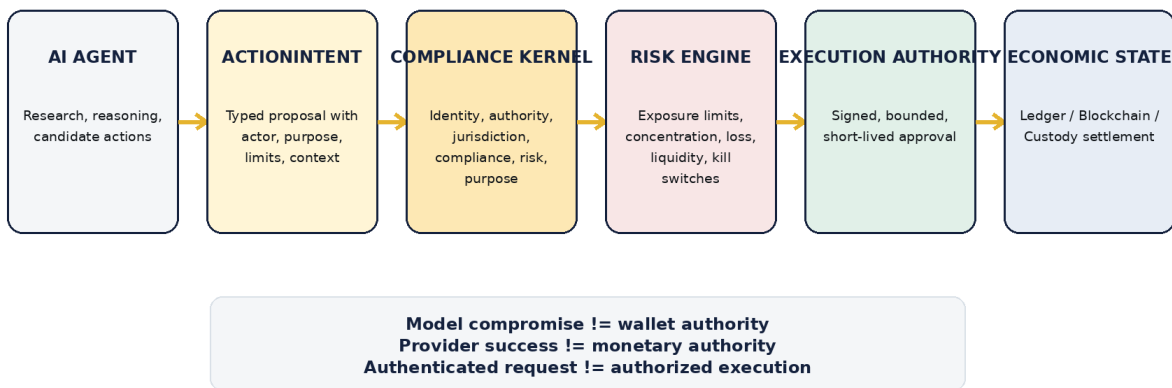


Figure 6. AI and user requests become typed ActionIntents; deterministic controls retain execution authority.

13.1 Proof-Carrying Financial Events

The Evidence Vault model records the ActionIntent together with identity state, KYC/AML/sanctions/fraud results, legal entity, jurisdiction, product/account eligibility, agreement/disclosure versions, agent capability, model version, risk result, policy version, approval decision, execution result, ledger entries, settlement state, timestamps and cryptographic references. The target question is mechanically simple: "Why was this action permitted at that exact time?" [S2]

TRANSACTION SECURITY

A secure transaction is not merely signed. It is signed by the right authority, for the right purpose, in the right jurisdiction, under the right risk and compliance state, and recorded in a canonical ledger with evidence.

14 | AI FINANCIAL AGENT SECURITY

AI Financial Agent Security

The security objective is to gain autonomous reasoning without granting autonomous financial sovereignty.

SunRey's Financial Agent architecture deliberately separates analysis from execution. Agents reason over customer-authorized context and emit ActionIntents. Capability tokens define allowed and forbidden actions, account classes, limits, jurisdiction, expiry and revocation. The deterministic Risk Engine outranks all trading AI, and models are expected to progress through research, testing, independent validation, shadow/paper/canary stages before production use. [S2]

14.1 Principal AI threats

Threat	Example	Control
Prompt injection	External content attempts to cause a privileged tool call	Treat model output as untrusted proposal; enforce tool capability in infrastructure
Data exfiltration	Agent attempts to expose private Vault context to an external model	Purpose firewall, data minimization, model/provider boundary
Hallucinated action	Model invents beneficiary, asset or permission state	Canonical data lookup plus deterministic validation
Privilege escalation	Agent requests action outside mandate	Capability token and Kernel denial
Compromised external model	Provider behavior becomes malicious or unavailable	Provider isolation, structured outputs, fallback/degraded mode, no execution credentials
Poisoned market data	Adversarial data biases recommendation	External data trust, corroboration, provenance, confidence state

AI SECURITY PRINCIPLE

A model compromise should degrade the quality of proposals, not silently grant custody, ledger, compliance, governance or wallet authority.

15 | INTERNAL ZERO-TRUST ARCHITECTURE

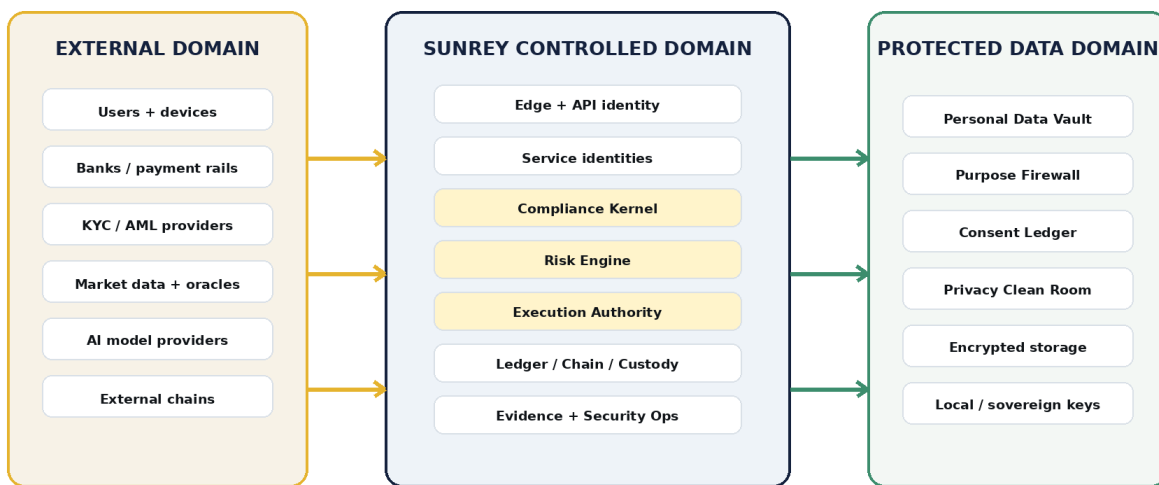
Internal Zero-Trust Architecture

Internal does not mean trusted.

SunRey's internal security baseline calls for zero trust, least privilege, strong service identity, mTLS, HSM/KMS-backed keys, MFA/passkeys, device binding, privileged access management, immutable audit logging, network segmentation, secret rotation and no secrets in source code. [S2][S6] This closely matches the resource-centric and identity-centric principles in NIST's zero-trust guidance. [E9]

Internal and External Cybersecurity Boundaries

External systems provide services and evidence; SunRey-owned controls retain economic authority.



Boundary rule: network location alone grants no trust. Identity, purpose and authorization are re-evaluated at consequential trans

Figure 7. External services, SunRey-controlled services and protected data remain separate trust domains.

15.1 Service-to-service trust

A service should not receive privilege because its IP address is "internal." Workload identity, explicit authorization, policy context and narrow service roles should be evaluated at each sensitive boundary. This matters in a microservice environment because lateral movement after a single workload compromise is often more damaging than the initial foothold.

16 | EXTERNAL PROVIDER AND API SECURITY

External Provider and API Security

External integrations should be replaceable, authenticated and unable to promote themselves into internal financial authority.

Banks, payment rails, card processors, KYC/AML vendors, custody providers, market-data feeds, oracles and AI providers sit behind SunRey-owned interfaces in the productization architecture. Provider runtime controls include lifecycle state, secret references, environment isolation, capability routing, health, timeouts, retries, circuit breakers, failover, kill switches and certification. [S6]

16.1 External trust rules

- Authenticate the provider and transport, but treat the provider response as evidence rather than financial authority.
- Normalize vendor-specific status into canonical SunRey states before product logic consumes it.
- Use scoped credentials and secret references; avoid embedding provider secrets in application source or frontend code.
- Verify inbound webhooks with signatures/timestamps/nonces where supported and make consumers idempotent.
- Fail closed for capabilities whose security, legal or provider evidence is missing.
- Maintain provider dependency, concentration-risk and exit-plan metadata for critical services.

EXTERNAL EVIDENCE RULE

Provider success may inform a decision. It should not mint native assets, bypass the Compliance Kernel, rewrite the Ledger or silently activate a regulated capability.

17 | HUMAN INFORMATION NETWORK AND DATA SECURITY

Human Information Network and Data Security

SunRey protects rights around information without turning the blockchain into a permanent public repository of sensitive source data.

SunRey's data architecture keeps sensitive source data off-chain and uses protected storage for user-linked information. The blockchain may carry suitable rights, consent, provenance, attestations, policy references and settlement evidence rather than raw personal records. [S4][S5]

17.1 Purpose-bound access

The Personal Data Vault, Consent Ledger and Purpose Firewall are security controls as much as privacy features. Every sensitive-data access should state a purpose, scope, requester, data categories, duration, jurisdiction, model class where relevant, compensation terms and revocation state. Possession of information never automatically implies permission to use it. [S1][S2]

17.2 Compute-to-data

For approved analytics and research, the architecture prefers computation-to-data over indiscriminate data export. A controlled clean-room environment can accept a query or model and return only permitted outputs. This reduces copying, limits data sprawl and provides a stronger place to enforce privacy thresholds, logging and purpose policy.

DATA MINIMIZATION

The strongest encryption strategy is still weaker than never moving unnecessary sensitive data into a broader trust domain in the first place.

18 | WALLET AND DIGITAL-ASSET CUSTODY SECURITY

Wallet and Digital-Asset Custody Security

Custody is a distinct authority domain with its own keys, approvals, limits and recovery processes.

The architecture calls for MPC and HSM-backed signing options, cold/warm/hot wallet patterns, withdrawal controls, rate limits, allowlists, transaction simulation, anomaly detection and strict separation of customer digital assets from corporate assets. [S2] The build architecture further describes canonical custody, signer providers, approval controls, finality handling and submission-ambiguity management. [S4]

18.1 Custody control objectives

- Private-key material should be non-exportable or tightly controlled where institutional infrastructure permits.
- Signing policy should include purpose, amount, destination, velocity, account class and approval state.
- Recovery should require stronger authority than ordinary transaction signing.
- Submission ambiguity should not cause duplicate transfer attempts; idempotency and finality tracking are required.
- Exchange balances should not become a competing custody or ledger truth source.

CUSTODY INVARIANT

The Exchange may reserve or request assets; canonical custody and chain/ledger state determine whether those assets actually exist and move.

19 | SUNREY EXCHANGE SECURITY

SunRey Exchange Security

A universal economic exchange introduces market-integrity, custody, data-rights and settlement risks beyond ordinary token trading.

SunRey Exchange is designed to host multiple market families, including digital assets, human-information rights, intelligence/compute and productive capacity. Orders may carry identity, rights, jurisdiction, purpose, consent, provenance, delivery conditions and policy state in addition to price and quantity. [S5]

19.1 Exchange security layers

Layer	Security requirement
Order entry	Authenticated actor, replay/idempotency controls, policy eligibility, market limits
Matching	Deterministic price-time or market-specific matching; no hidden balance authority
Custody	Pre-trade reservation and canonical custody linkage
Settlement	Atomic or carefully orchestrated DVP; explicit failure/recovery state
Surveillance	Market-abuse detectors, restrictions, concentration/velocity checks, kill switches
Data-rights markets	Purpose, consent, privacy threshold and delivery evidence
Operational controls	Pause by market, asset, customer, provider or jurisdiction without disabling the entire platform

The cryptographic problem is therefore only one component. The Exchange must also defend against logically valid but unauthorized market actions, stale or manipulated external state, inconsistent settlement, compromised custody, duplicated execution and policy drift.

20 | ORACLE AND EXTERNAL-DATA SECURITY

Oracle and External-Data Security

External information may inform economic state without becoming monetary authority.

SunRey relies on external observations for markets, productive capacity, asset state, delivery and economic facts. The security problem is not merely authenticating the API; it is establishing whether an observation is authoritative enough, fresh enough and independent enough to support a specific action.

20.1 External data trust

- Source identity and authorization
- Freshness and timestamp policy
- Provenance and transformation lineage
- Cross-source corroboration and source independence
- Outlier/anomaly handling
- Provider health and degraded-mode state
- Canonical event identity and anti-double-counting
- Explicit separation between evidence, valuation and monetary authority

ORACLE RULE

A signed oracle message can prove what an oracle said. It does not prove that the statement is true, sufficiently corroborated, non-duplicative or authorized to alter native monetary supply.

21 | INTEROPERABILITY AND BRIDGE SECURITY

Interoperability and Bridge Security

External chains may interoperate without becoming the native source of truth for SunRey economic state.

The SunRey architecture keeps external chains outside the authoritative base layer and treats relayers as untrusted. Source documents explicitly prefer proof-oriented interoperability over trusted-multisig bridge roots and reject wrapped-fiat shortcuts as an authority substitute. [S4][S5]

21.1 Bridge control framework

Risk	Control direction
Relayer compromise	Verify proofs/finality rather than trust relayer assertions
Replay	Bind messages to source chain, destination, nonce/sequence, asset and domain
External reorganization	Model finality assumptions and delay/limit settlement appropriately
Wrapped-asset insolvency	Separate representation risk from native-asset truth; require reserve/custody evidence where applicable
Bridge key compromise	Threshold governance, rate limits, emergency suspension, narrow key scope
Protocol bug	Allowlist supported pathways, bounded limits, staged rollout and kill switches

INTEROP PRINCIPLE

Interoperability should import verifiable state or authorized economic relationships - not import another network's security assumptions as SunRey's root of trust.

22 | CLOUD AND INFRASTRUCTURE SECURITY

Cloud and Infrastructure Security

Production infrastructure should preserve one canonical state path, strict network exposure and environment isolation.

SunRey's deployment architecture has used multiple infrastructure environments during productization, but the security rule remains one authoritative state for each domain. Persistent databases, blockchain nodes, custody services and economic state should not be duplicated in ways that create competing financial truth. [S6]

22.1 Infrastructure control classes

- Cloud/hosting accounts protected by strong operator identity, MFA and least privilege.
- Databases and internal services not directly exposed to the public Internet unless specifically required.
- Reverse proxy / edge services enforce HTTPS, request limits and intended origin boundaries.
- Infrastructure-as-code and versioned configuration reduce undocumented drift.
- Secrets stored outside source code and rotated through managed systems.
- Environment separation prevents sandbox/testnet credentials or fixtures from silently becoming production authority.
- Backups encrypted, access-separated and regularly restore-tested.

PUBLICATION SAFETY

This paper intentionally omits IP addresses, secret names/values, private ports, administrative endpoints and detailed firewall configuration.

23 | SOFTWARE SUPPLY-CHAIN SECURITY

Software Supply-Chain Security

The release pipeline is itself part of the financial-security perimeter.

SunRey build reports describe architecture linting, TypeScript and Rust qualification, secret scans, formal/fuzz/adversarial testing, OCI artifact construction, SBOM/provenance and release-candidate gates. [S4][S6] These practices are consistent with the principle behind NIST's Secure Software Development Framework: secure development practices should be integrated into the SDLC rather than added only after release. [E10]

23.1 Source-to-deployment chain

Stage	Security requirement
Source	Protected main branch, review, code ownership, no secrets
Dependencies	Lockfiles, provenance awareness, vulnerability monitoring
Build	Clean/reproducible process where practical, architecture and type/lint gates
Test	Unit/integration/E2E, financial invariants, security, fuzz/adversarial checks
Artifact	Immutable digest, SBOM, provenance/attestation, exact release identity
Promotion	Environment-specific approvals and external production gates
Deployment	Pinned release; no floating "latest" as authority
Rollback	Known-good artifact and state-compatible recovery procedure

CODE EXISTS != CAPABILITY AUTHORIZED

A production-capable code path can remain disabled until legal, provider, security, governance and operational evidence exists.

24 | FORMAL, FUZZ AND ADVERSARIAL ASSURANCE

Formal, Fuzz and Adversarial Assurance

Testing should target both implementation defects and failures of authority boundaries.

Source build reports describe formal models, fuzzing, adversarial-range testing, signer safety tests and extensive regression suites. [S4] The publication claim should remain bounded: internal tests are valuable engineering evidence, but they are not substitutes for independent review of cryptography, consensus, custody or production infrastructure.

24.1 Assurance portfolio

- Unit tests for deterministic functions and cryptographic adapters.
- Property tests for invariants such as balanced ledgers, idempotency and monotonic authorization.
- Fuzzing of parsers, transaction envelopes, network messages and state-transition surfaces.
- Adversarial tests for double-signing, replay, privilege escalation, invalid state transitions, provider failures and malformed evidence.
- Concurrency/crash testing for settlement, jobs, wallet operations and persistence.
- Chaos tests for provider loss, cell isolation, database/network disruption and degraded modes.
- Independent penetration and protocol reviews before live activation.

ASSURANCE PRINCIPLE

The relevant question is not "How many tests exist?" but "Which failure modes and authority invariants have been exercised, under what assumptions, and with what independent evidence?"

25 | OBSERVABILITY AND SECURITY OPERATIONS

Observability and Security Operations

A secure economic system must be able to detect divergence and reconstruct cause without giving the monitoring plane execution authority.

SunRey's operations architecture calls for metrics, logs, traces, SLIs/SLOs, alerts, incident records, control-room views, runbooks and provider/financial/security operations APIs. [S6] These signals should be correlated around stable transaction, user, agent, model, provider, release and evidence identifiers so that operational events can be traced across domains.

25.1 High-value detections

- Repeated authentication and step-up failures
- Unexpected privilege or role changes
- Abnormal transaction velocity or destination changes
- Ledger reconciliation breaks
- Validator or signer anomalies
- Custody approval/signing failures
- Market surveillance events
- Provider timeout/error pattern shifts
- AI tool-call denials or unusual proposal patterns
- Backup/restore failures
- Secret-access anomalies
- Unexpected production-gate or configuration changes

OBSERVABILITY BOUNDARY

Monitoring may reveal, classify and recommend. It should not silently become a money-moving, governance or protocol-upgrade authority.

26 | INCIDENT RESPONSE, RECOVERY AND RESILIENCE

Incident Response, Recovery and Resilience

Recovery must restore trusted authority, not merely restore service availability.

SunRey security guidance includes multi-region capability, database replication, point-in-time recovery, immutable backups, backup verification, recovery exercises, provider-dependency mapping and independent kill switches across trading, payments, external transfers, crypto withdrawals, Exchange, data markets, agents, customers and jurisdictions. [S2]

26.1 Incident recovery sequence

1. Contain the affected trust domain and activate the narrowest appropriate kill switch.
2. Preserve evidence and establish the last known trustworthy state.
3. Revoke or rotate compromised credentials/keys and quarantine affected providers or services.
4. Restore from verified state or replay canonical ledgers/events as designed.
5. Reconcile financial and blockchain state before reopening execution.
6. Requalify the repaired environment and document accepted residual risk.
7. Only then reactivate capability through the appropriate human/governance process.

RESILIENCE PRINCIPLE

Availability without trustworthy state is not recovery. Financial and blockchain systems must restore both service and confidence in canonical economic state.

27 | SECURITY GOVERNANCE AND SEPARATION OF DUTIES

Security Governance and Separation of Duties

No single AI, administrator, signer or deployment system should become protocol governance.

SunRey architecture separates protocol governance, security operations, infrastructure, compliance, custody, financial operations, software release, validator operations and AI/model governance. Source documentation calls for explicit protocol versions and activation coordinates with human threshold authority. [S4]

27.1 Authority domains

Authority	May authorize	Should not independently authorize
Security operations	Containment, credential revocation, security response	Monetary policy or arbitrary fund movement
Compliance / policy	Eligibility, jurisdiction and regulatory conditions	Custody signing or protocol upgrades
Custody	Approved transaction signing within policy	Product eligibility or model mandates
Protocol governance	Version activation, validator/economic policy decisions	Private customer financial operations
Software release	Artifact production and deployment candidate	Live activation absent required gates
AI/model governance	Model approval, purpose, release state, monitoring	Ledger writes or governance signatures
Financial operations	Treasury/settlement workflows under controls	Bypass of canonical ledger or compliance decisions

THRESHOLD AUTHORITY

High-impact changes should require multiple independent evidence and approval domains, not merely multiple signatures controlled by the same operational pathway.

28 | POST-QUANTUM MIGRATION ROADMAP

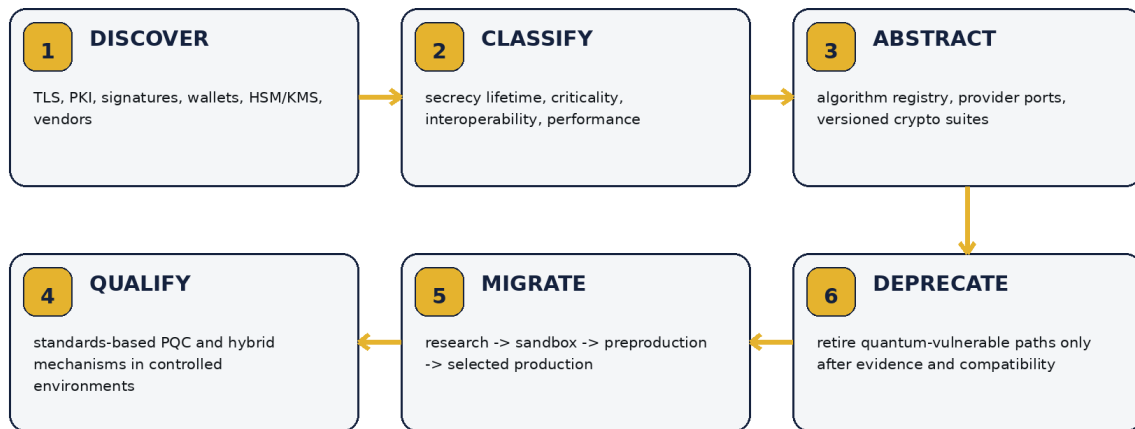
Post-Quantum Migration Roadmap

The transition should be managed as a security program with inventory, classification, qualification and controlled activation.

CISA/NSA/NIST guidance emphasizes starting with cryptographic discovery and inventory because organizations often lack visibility into where public-key cryptography is embedded. CISA has continued this work with automated PQC discovery and inventory strategy. [E6][E7] SunRey's migration roadmap follows the same operational logic.

Post-Quantum Migration Architecture

SunRey treats PQC as a governed transition across cryptographic dependencies, not a one-time algorithm swap.



Target property: preserve confidentiality, authenticity, authorization and operational continuity while cryptographic assumptions evolve.

Figure 8. Six-phase post-quantum migration roadmap.

28.1 Phase I - Cryptographic discovery

Inventory TLS, certificates, PKI, API authentication, workload identity, blockchain signatures, wallet/custody signing, validator identity, governance signatures, database encryption, backups, code/release signing, HSM/KMS dependencies and third-party providers.

28.2 Phase II - Classification

Prioritize by secrecy lifetime, authentication criticality, asset value, migration complexity, interoperability, hardware/provider support and performance constraints.

28.3 Phase III - Crypto-agility enforcement

Remove unnecessary algorithm coupling; centralize algorithm policy and provider interfaces; version protocol suites; make deprecation and fallback observable.

28.4 Phase IV - Hybrid PQC qualification

Evaluate standards-based PQC/hybrid mechanisms in isolated environments. Measure key/signature size, latency, CPU, memory, bandwidth, block-size impact, validator throughput, mobile performance, HSM/KMS/custody support and failure behavior.

28.5 Phase V - Controlled migration

Promote selected paths through research -> sandbox/testnet -> preproduction -> controlled production only when interoperability, operations, evidence and independent review are sufficient.

28.6 Phase VI - Legacy deprecation

Retire quantum-vulnerable mechanisms only after dependent systems are migrated or explicitly risk-accepted and rollback/recovery paths are understood.

29 | POST-QUANTUM MATURITY MODEL

Post-Quantum Maturity Model

A maturity model is more defensible than a binary claim that the platform is or is not "post-quantum."

Level	Definition	Evidence threshold
0 - Unknown	Cryptographic inventory incomplete	Discovery gaps documented
1 - Crypto-agile architecture	Algorithms/providers can be versioned and changed	Registry, interfaces, policy and tests
2 - PQC-capable components validated	Selected PQ primitives work in controlled components	Vectors, interoperability/performance tests
3 - Hybrid PQC qualified	Hybrid mechanisms operate in qualified environments	End-to-end sandbox/preproduction evidence
4 - Selected production domains PQ-protected	Specific live paths use qualified PQ/hybrid mechanisms	Production configuration + monitoring + assurance
5 - Broad migration with residual-risk register	Most priority quantum-vulnerable dependencies retired or bounded	Enterprise inventory, independent verification, exceptions and deprecation plan

Based on current source material, SunRey can credibly describe its design as ****crypto-agile with post-quantum migration foundations**** and report specific sandbox/testnet component work when reverified. The architecture does not support a blanket Level 4 or Level 5 claim across all production domains. [S4]

RECOMMENDED PUBLIC LANGUAGE

"SunRey is designed as a crypto-agile financial and blockchain platform with a standards-based post-quantum migration architecture."

30 | EXTERNAL ASSURANCE AND PRODUCTION GATES

External Assurance and Production Gates

Internal engineering evidence is necessary; independent evidence is required for credible production claims.

SunRey's own productization audit correctly identifies security audit, penetration test/retest, protocol review, HSM/KMS attestations and production key ceremonies as external launch gates. It also distinguishes current architecture from production provider, validator, legal and operational inputs. [S6]

30.1 External assurance package

- Independent application/API penetration test and remediation retest.
- Blockchain consensus/protocol and economic-state review.
- Cryptographic design and implementation review, including hybrid/PQC configuration where used.
- HSM/KMS/custody provider evidence and key-management assessment.
- Cloud/infrastructure and IAM assessment.
- Supply-chain/release provenance review.
- Privacy/HIN architecture review and data-flow assessment.
- Incident-response and recovery exercise with evidence.
- Validator operator security assessment and signing-process review.
- Production cryptographic inventory with unresolved dependency register.

ASSURANCE BOUNDARY

A green CI pipeline, a testnet or an internal red-team suite does not equal independent certification. Public claims should name the assessed scope, version, environment and date.

31 | STRATEGIC DIFFERENTIATION

Strategic Differentiation

The potentially distinctive security idea is the integration of cryptographic trust with economic authority boundaries.

None of the individual ingredients - TLS, HSMS, KMS, zero trust, PQC, BFT consensus, policy engines, evidence stores or AI guardrails - should be claimed as individually novel. The potentially differentiating architecture lies in how these controls are composed around a sovereign economic system.

31.1 Integrated security fabric

- Crypto-agile protocol and service architecture with explicit PQ migration state.
- AI proposal/execution separation enforced by deterministic authorization.
- Proof of Permission spanning identity, authority, jurisdiction, compliance, risk and purpose.
- Native blockchain state that recognizes rights, consent, provenance, AI/machine identity and policy context.
- Protected off-chain human information with rights/evidence anchored to the economic system.
- Provider evidence that is useful but not self-authorizing.
- Canonical ledger/chain/custody ownership boundaries preventing shadow financial truth.
- Evidence-producing execution and recovery, linking security operations to reconstructable economic events.
- Governance separation so keys, releases and models cannot silently become monetary or protocol authority.

POTENTIAL INNOVATION THESIS

A sovereign economic infrastructure in which cryptography, identity, policy, AI authority, economic execution and evidence operate as one security fabric. Formal novelty or patentability would require separate prior-art analysis.

32 | CONCLUSION

Conclusion

Post-quantum security is not an algorithm swap; it is the preservation of trustworthy economic authority as technology evolves.

The central security problem of an AI-native economic system is not only whether an attacker can read encrypted information. It is whether any compromised component can acquire authority it was never intended to possess. SunRey's architecture addresses this by separating reasoning from permission, permission from execution, execution from canonical state, and canonical state from the evidence used to reconstruct it.

Post-quantum cryptography strengthens that architecture by preparing the public-key trust layer for a future change in cryptographic assumptions. NIST's standardized ML-KEM, ML-DSA and SLH-DSA, modern KEM guidance, 2026 crypto-agility guidance and IETF hybrid TLS standardization now provide a substantially stronger external foundation for migration planning than existed only a few years ago. [E1-E5][E8]

FINAL THESIS

Post-quantum security is the ability of an economic system to preserve confidentiality, authenticity, authorization and monetary integrity as cryptographic assumptions, infrastructure and autonomous agents evolve.

SunRey's strongest security claim is therefore not that it has eliminated cyber risk. It is that the architecture is being designed so security decisions remain explicit, bounded, replaceable, evidence-producing and independently verifiable - from user identity and AI proposals through blockchain consensus, financial execution, key management and operational recovery.

A | GLOSSARY

Glossary

Term	Definition
ActionIntent	Structured request or proposal describing a consequential action before authorization.
Compliance Kernel	Deterministic control plane evaluating identity, authority, jurisdiction, compliance, risk and purpose.
Execution Authority	Signed, bounded and short-lived authorization accepted by controlled execution infrastructure.
Crypto agility	Capability to replace/adapt cryptographic algorithms while preserving security and operations.
KEM	Key-encapsulation mechanism used to establish a shared secret over a public channel.
ML-KEM	NIST-standardized module-lattice KEM in FIPS 203.
ML-DSA	NIST-standardized module-lattice digital signature algorithm in FIPS 204.
SLH-DSA	NIST-standardized stateless hash-based digital signature algorithm in FIPS 205.
Hybrid PQ/traditional	Use of both a post-quantum and a conventional mechanism so security can remain robust during migration.
KeyProvider	SunRey abstraction separating application logic from specific key-management/signing implementations.
Sovereign Cell	Regional deployment boundary with local data, keys, policies, ledgers/legal entities and recovery infrastructure.
Evidence Vault	Tamper-evident/WORM-oriented evidence store supporting reconstruction and audit.
HIN	Human Information Network - rights-governed architecture for authorized human information and contribution.
Canonical state	Authoritative ledger/blockchain/custody state that other systems may reference but not duplicate as competing truth.

B | REFERENCES

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REFERENCE NOTE

External references are used for standards context and fact-checking. They do not imply NIST, CISA, NSA or IETF endorsement of SunRey.